

Sales Data Analysis System

```
#import Libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
#Data Frame
data={
    "Products":["Mobiles","Smartwatches","Laptop","Tablet","Air Conditionor","Smart Lights"],
    "Price":[30000,3000,50000,15000,23000,1500],
    "Sold":[10,15,8,10,None,5]
}
df=pd.DataFrame(data)
print("Product Data:\n")
print(df)
```

Product Data:

	Products	Price	Sold
0	Mobiles	30000	10.0
1	Smartwatches	3000	15.0
2	Laptop	50000	8.0
3	Tablet	15000	10.0
4	Air Conditionor	23000	NaN
5	Smart Lights	1500	5.0

```
#Sum of Price
total_sales=np.array(df["Price"].sum())
print("Total_Sales:",total_sales)
#Sum of Sold
total_sold=np.array(df["Sold"].sum())
print("Total Sold:",total_sold)
#Adding total_sales
df["total_sales"]=df["Price"]*df["Sold"]
```

Total_Sales: 122500
Total Sold: 48.0

```
#Adding values in None
df["Sold"]=df["Sold"].fillna(10)
print(df)
```

	Products	Price	Sold	total_sales
0	Mobiles	30000	10.0	300000.0
1	Smartwatches	3000	15.0	45000.0
2	Laptop	50000	8.0	400000.0
3	Tablet	15000	10.0	150000.0
4	Air Conditionor	23000	10.0	NaN
5	Smart Lights	1500	5.0	7500.0

```
#calculate total_sales after adding value
df["total_sales"]=df["Price"]*df["Sold"]
print(df)
```

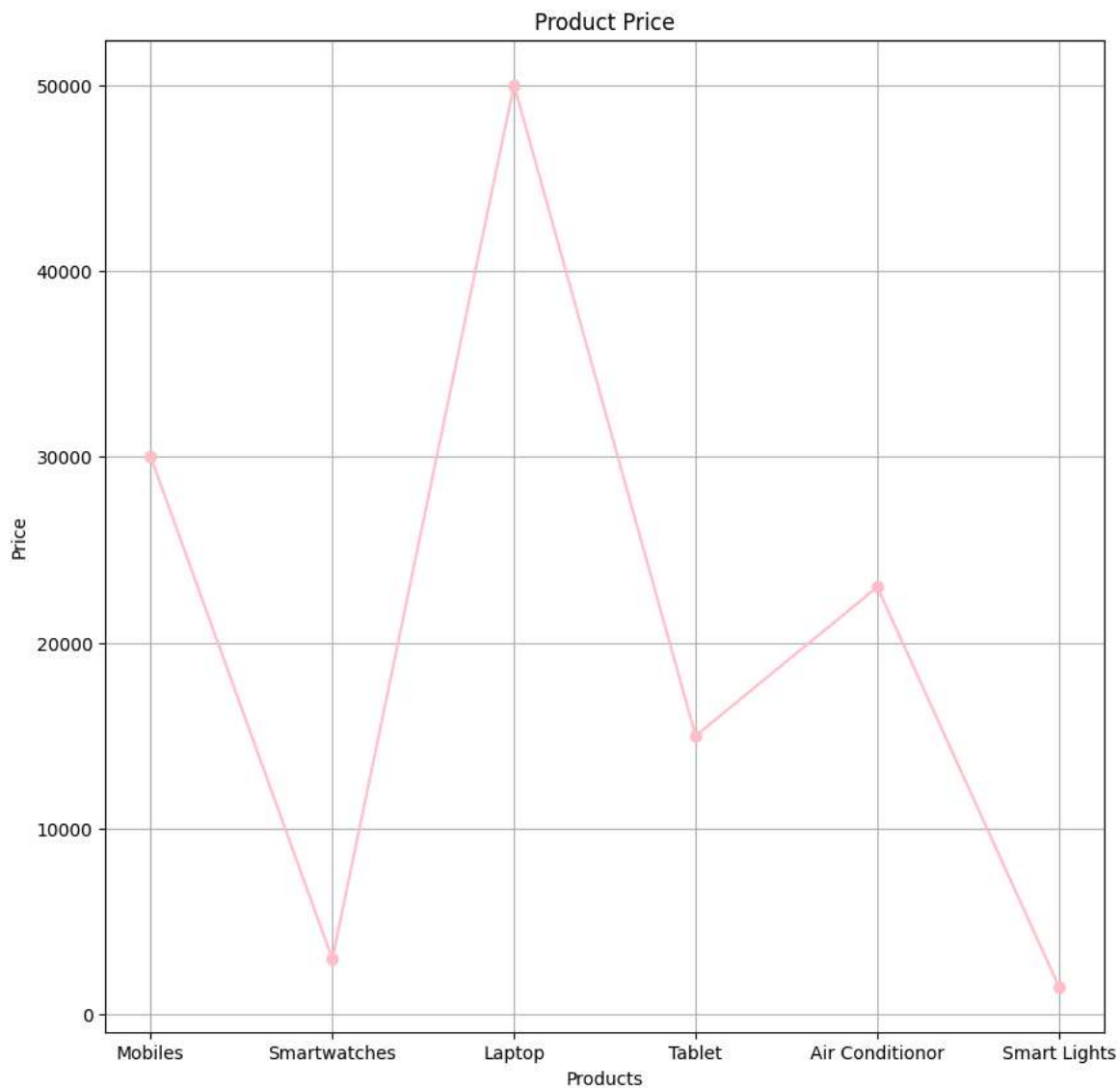
	Products	Price	Sold	total_sales
0	Mobiles	30000	10.0	300000.0
1	Smartwatches	3000	15.0	45000.0
2	Laptop	50000	8.0	400000.0
3	Tablet	15000	10.0	150000.0
4	Air Conditionor	23000	10.0	230000.0
5	Smart Lights	1500	5.0	7500.0

```
#Total income
a=df["total_sales"].sum()
print("total income:",a)
```

total income: 1132500.0

```
#Line chart
import matplotlib.pyplot as plt
plt.figure(figsize=(10,10))
plt.plot(df["Products"],df["Price"],marker="o",color="pink")
```

```
plt.title("Product Price")
plt.xlabel("Products")
plt.ylabel("Price")
plt.grid(True)
plt.show()
```



```
#Bar chart
fig, ax = plt.subplots(figsize=(10, 5))
bars = ax.bar(df["Products"], df["total_sales"], color='#1976D2', edgecolor='white', width=0.6)
for bar in bars:
    height = bar.get_height()
    ax.text(bar.get_x() + bar.get_width()/2, height + 50,
            f'₹{height:}', ha='center', va='bottom', fontsize=10)
ax.set_title('Revenue of Products', fontsize=15, pad=15)
ax.set_xlabel('Price')
ax.set_ylabel('Sold')
ax.set_ylim(0, 600000)
plt.show()
```

